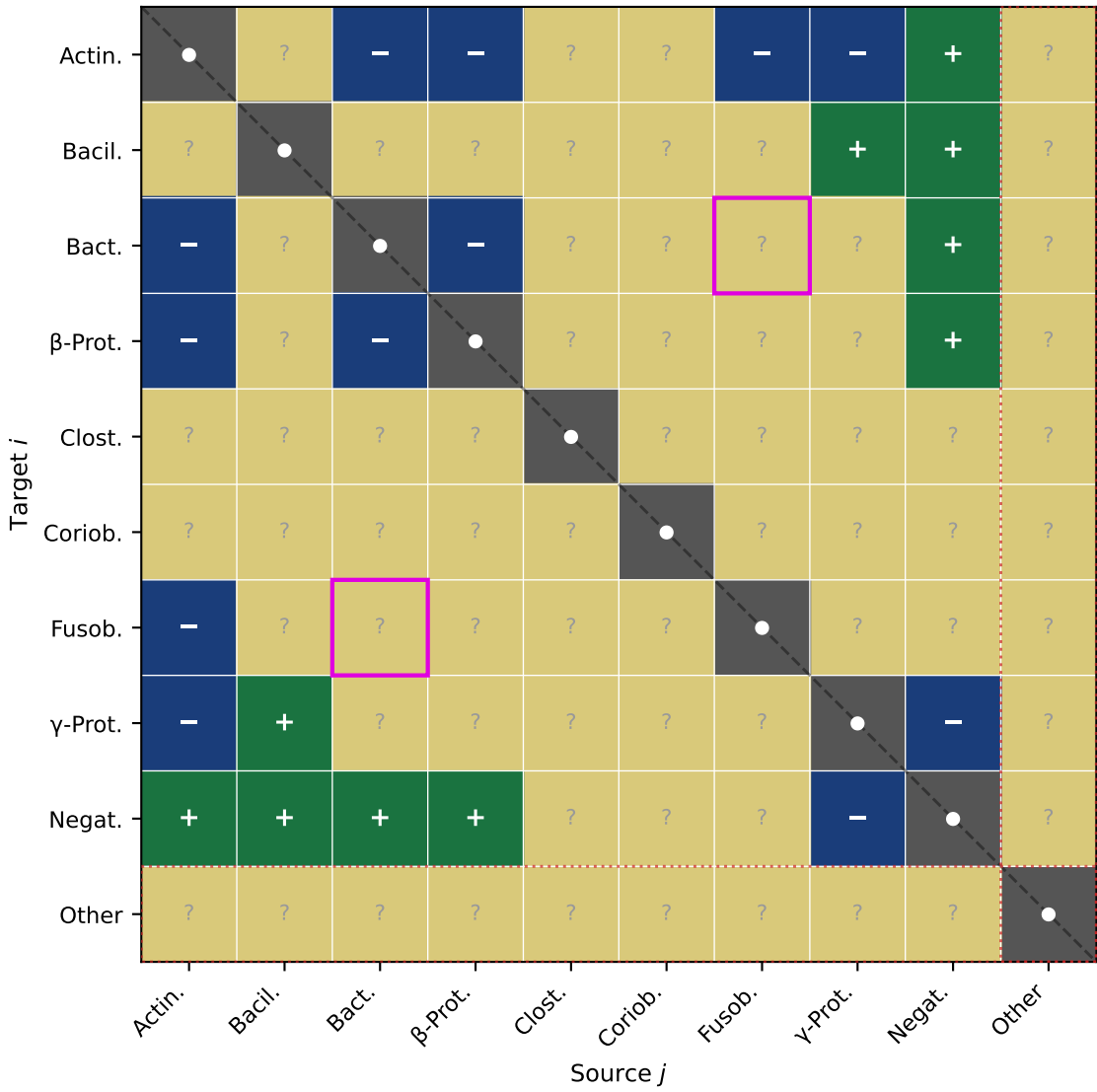


L1+L2

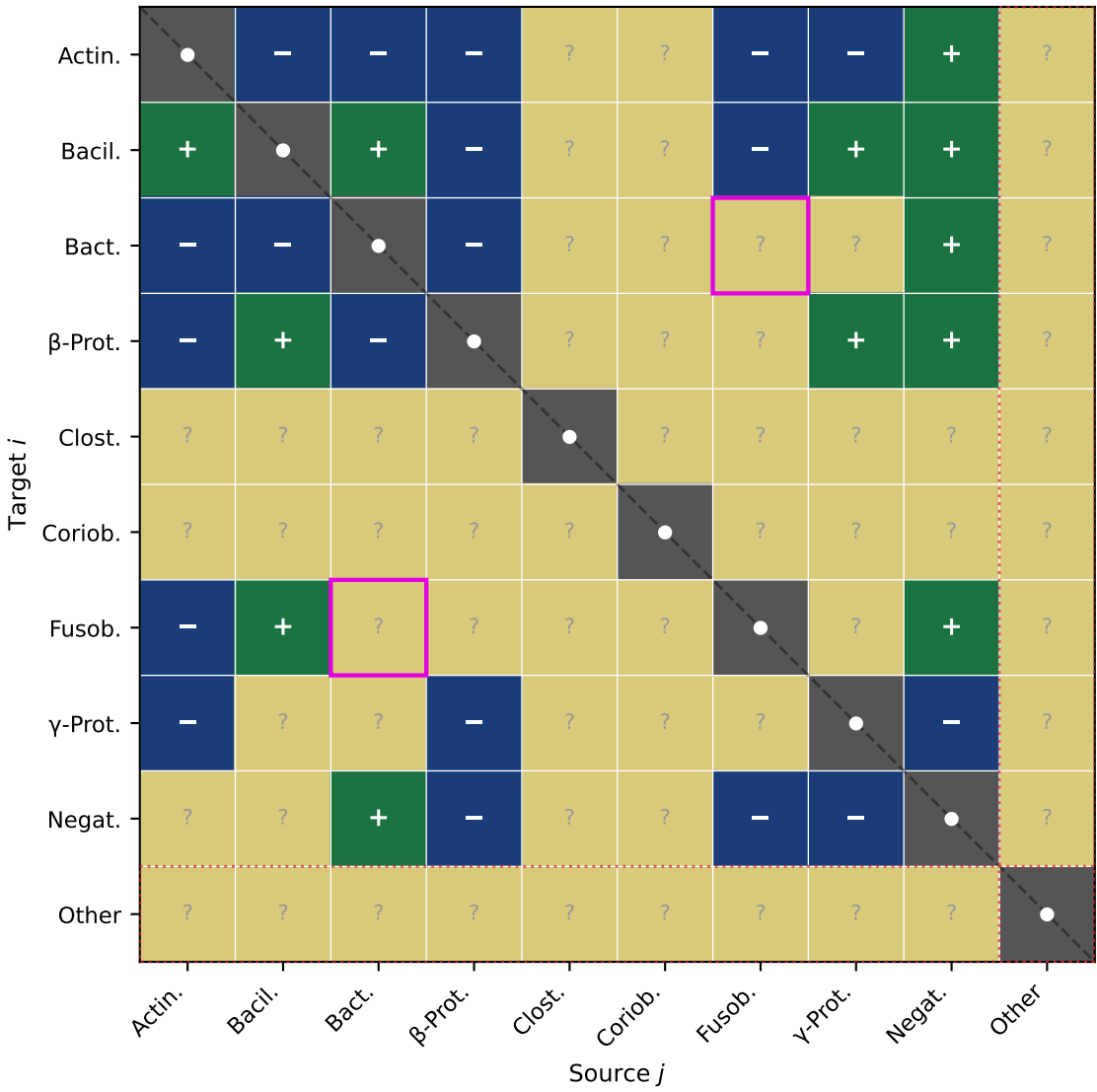
Symmetric ($A_{ij} = A_{ji}$, Hamilton)

A1 L1+L2 symmetric
11/45 constrained (+:5 -:6)



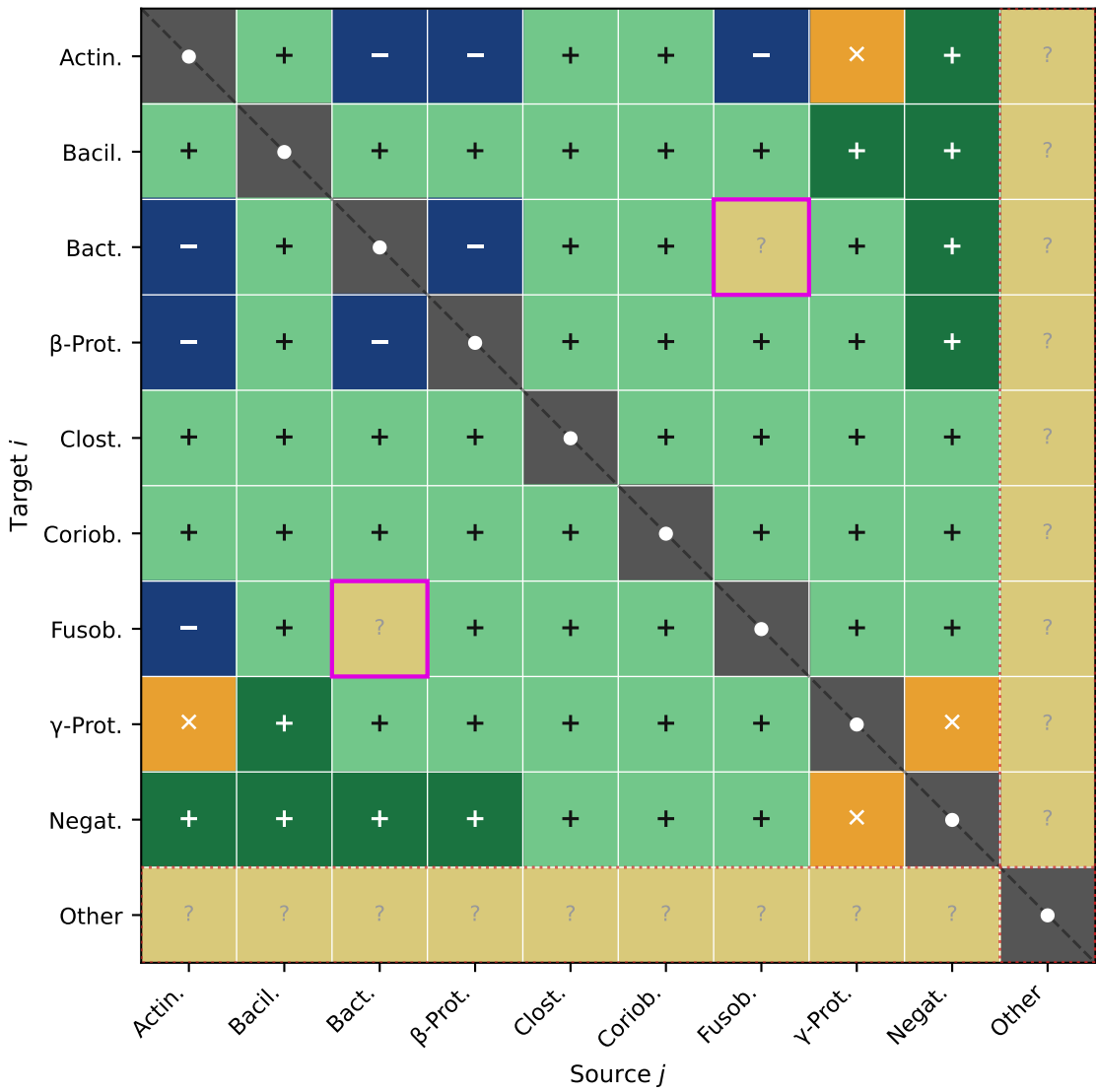
Directed / Asymmetric ($A_{ij} \neq A_{ji}$, gLV)

A2 L1+L2 directed
31/90 constrained (+:12 -:19)

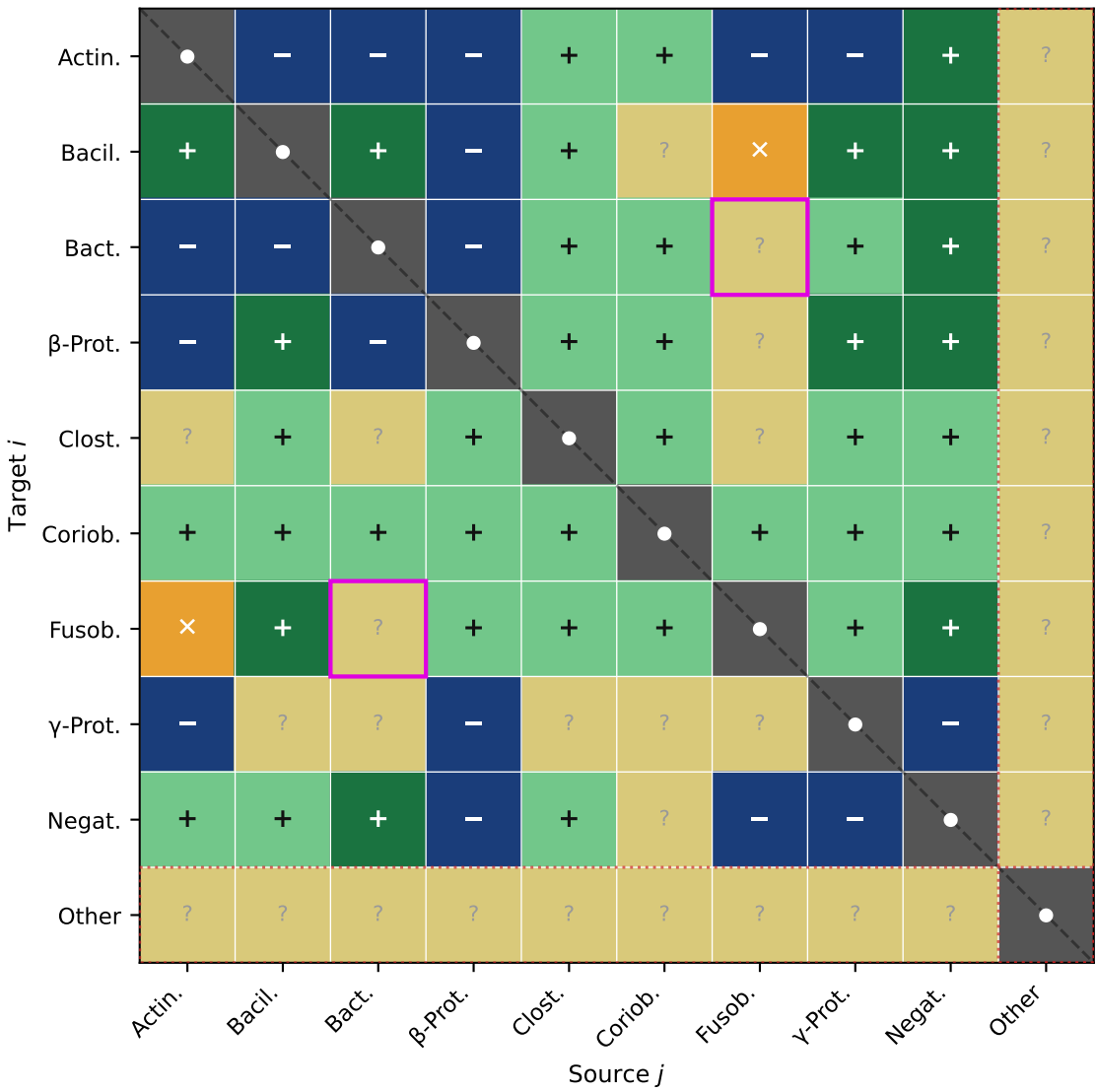


L1+L2 + AGORA

B1 L1+L2 + AGORA symmetric
33/45 AGORA+:24 cancelled(x):2 free:10

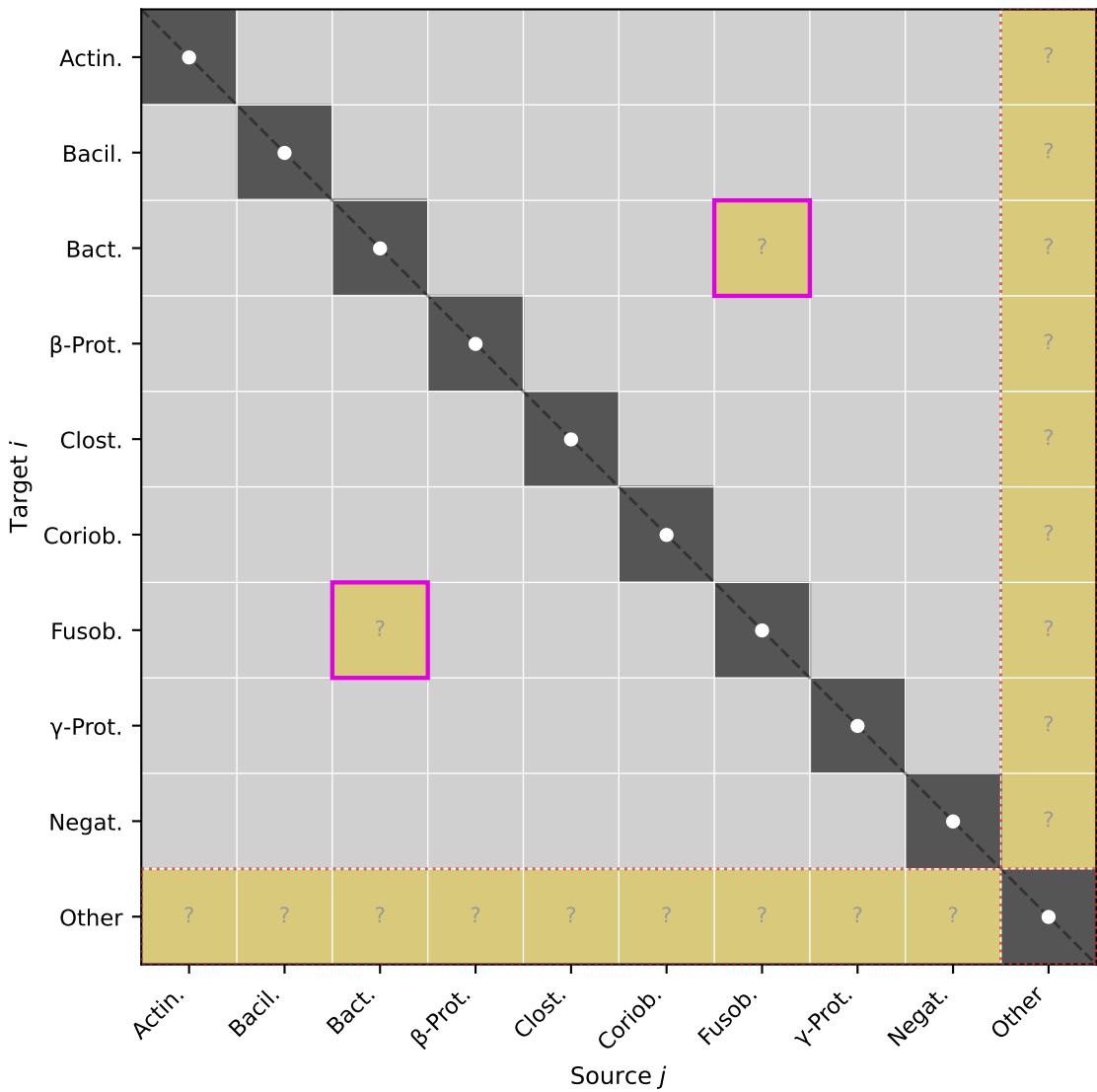


B2 L1+L2 + AGORA directed
57/90 AGORA+:28 cancelled:2 free:31

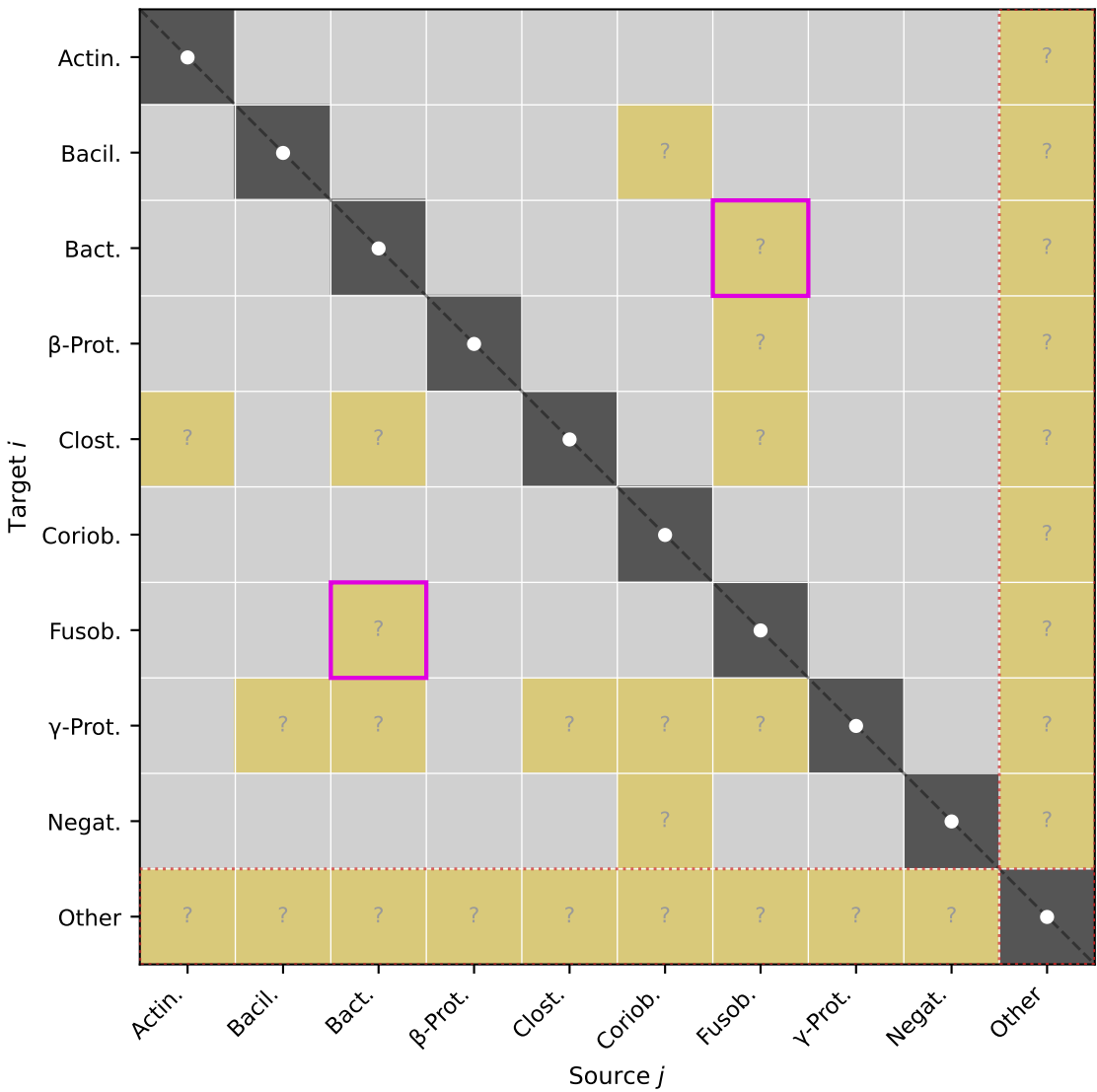


Unconstrained

C1 Unconstrained symmetric
10/45 pairs (? = no sign prior)



C2 Unconstrained directed
31/90 directed pairs (? = no sign prior)



L1+L2 positive (+) AGORA added (+) L1+L2 cancelled by AGORA (x) Constrained (dimmed in Row C) † Other: excluded from AGORA
L1+L2 negative (-) AGORA added (-) Unconstrained (?) Self-limitation (diagonal) ■ Bact-Fusob: no metabolic signal